

Formal Verification: DeFi Fee Model

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Abstract

We formalize the maker-taker fee model with volume-based tiers. The key properties are: maker fees are always $\leq$ taker fees (incentivizing liquidity provision), higher volume yields lower fees, and validator revenue is bounded.

1 Introduction

The fee model incentivizes liquidity provision (lower maker fees) and rewards high-volume participants (volume tiers). The permissionless fee schedule is on-chain and transparent.

2 Definitions

Definition 1 (FeeTier). *A fee tier: minimum volume, maker basis points, taker basis points.*

3 Theorems and Axioms

Theorem 2 (maker_le_taker). *Maker fee $\leq$ taker fee for any tier where makerBps $\leq$ takerBps.*

Proof. By monotonicity of integer division. □

Theorem 3 (top_tier_zero). *Top tier (\$100M+) has zero maker fee.*

Theorem 4 (volume_discount). *Higher volume tiers have lower taker fees.*

Theorem 5 (four_tiers). *The fee schedule has exactly 4 tiers.*

Theorem 6 (validator_bounded). *Validator share of fees never exceeds total fees.*

Proof. By division bound. □

4 Lean Source

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DeFi Fee Model

Maker-taker with volume tiers. Permissionless.

No regulatory fee caps (pure DeFi).

Maps to:

- *lux/exchange*: fee calculation
- *lux/node*: fee distribution to validators

Properties:

- *Makers takers* (incentivize liquidity)
- *Volume tiers*: more volume $\rightarrow$ lower fees
- *Validator revenue*: portion of fees to validators
- *Permissionless*: fee schedule is on-chain and transparent

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```
import Mathlib.Data.Nat.Defs
import Mathlib.Tactic
```

```
namespace DeFi.FeeModel
```

```
/-- Fee tier -/
```

```
structure FeeTier where
  minVolume : Nat
  makerBps : Nat
  takerBps : Nat
  deriving DecidableEq, Repr
```

```
/-- Standard DeFi tiers -/
```

```
def tiers : List FeeTier :=
[ 0,      15, 30      -- Tier 1
, 1000000, 10, 25     -- Tier 2: $1M+
, 10000000, 5, 15     -- Tier 3: $10M+
, 100000000, 0, 10    -- Tier 4: $100M+ (zero maker)
]
```

```
/-- Calculate fee -/
```

```
def fee (tier : FeeTier) (isMaker : Bool) (value : Nat) : Nat :=
  value * (if isMaker then tier.makerBps else tier.takerBps) / 10000
```

```
/-- MAKER    MAKER    -/
```

```
theorem maker_le_taker (t : FeeTier) (v : Nat) (h : t.makerBps <= t.takerBps) :
  fee t true v <= fee t false v := by
  simp [fee]; exact Nat.div_le_div_right (Nat.mul_le_mul_left v h)
```

```
/-- TOP TIER ZERO MAKER -/
```

```
theorem top_tier_zero : (tiers[3]!).makerBps = 0 := by simp [tiers]
```

```
/-- VOLUME DISCOUNT -/
```

```
theorem volume_discount : (tiers[0]!).takerBps > (tiers[3]!).takerBps := by
  simp [tiers]
```

```
/-- 4 TIERS -/
```

```
theorem four_tiers : tiers.length = 4 := rfl
```

```
/-- VALIDATOR SHARE: Portion of fees goes to validators who process trades -/
```

```
def validatorShare (totalFee : Nat) (validatorPct : Nat) : Nat :=
  totalFee * validatorPct / 100
```

```
theorem validator_bounded (totalFee pct : Nat) (h : pct < 100) :  
  validatorShare totalFee pct < totalFee := by  
  simp [validatorShare]; exact Nat.div_le_of_le_mul (by omega)  
  
end DeFi.FeeModel
```

5 References

Source code: <https://github.com/luxfi/proofs/blob/main/lean/DeFi/FeeModel.lean>

White papers: <https://papers.lux.network>

Related white papers:

- [lux-light-speed-dex.tex](#)